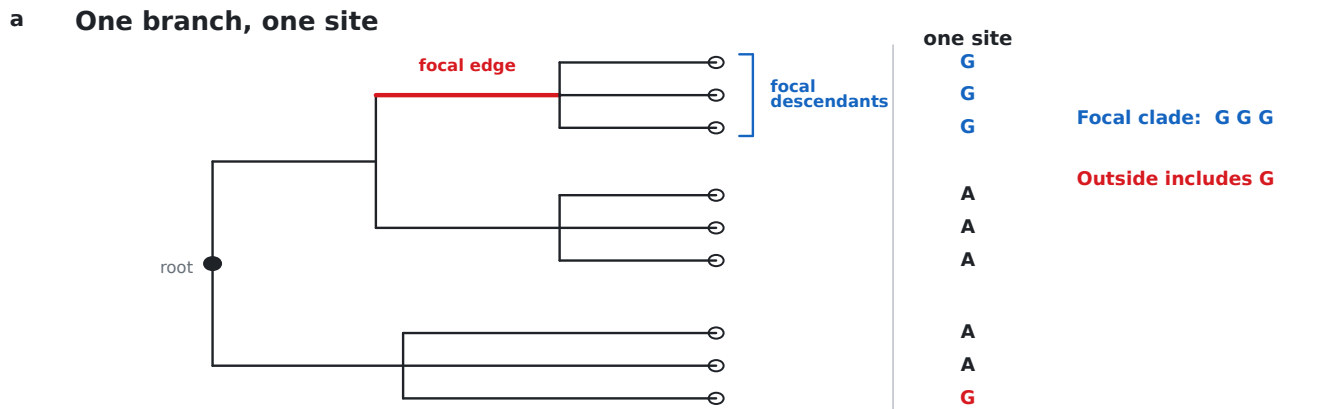




b Two questions, different answers

OBSERVED

Clade-exclusive?

focal	G	G	G			
outside	A	A	A	A	A	G

NO not fixed-exclusive
G occurs outside the clade

kept separate

INFERRED

Substitution on the focal edge?

YES unambiguous A→G
Only optimal focal-edge pair: A→G

c Retain all optimal focal-edge state pairs

Optimal parent→child pair(s)	BRANCHSNV reports
A→G only	Unambiguous change
A→G or C→G	State ambiguity
A→G or G→G	Placement ambiguity
G→G only	No change

No arbitrary tie-breaking
All globally optimal focal-edge state pairs are retained.

DESIGN SAFEGUARDS

- Explicit root**
defines direction
- Exact descendant set**
defines branch
- Exact taxon labels**
define correspondence
- Complete optimal set**
retains uncertainty
- Deterministic provenance**
supports auditability